

MINGZHE LI

Financial Economics, Household Finance, Continuous Time
Shihezi University ◊ mingzheli@stu.shzu.edu.cn

WORKING PAPERS

On the Monotonicity and Concavity of Effective Risk Aversion (2026) *Available at SSRN 6911380*

Abstract: Apparent risk aversion has been extensively studied in the savings problem, but the corresponding portfolio choice object—effective risk aversion, which directly determines the risky portfolio share—has not been characterized for heterogeneous investors. For HARA utility functions with $k > 1$, apparent prudence is greater than k times apparent risk aversion if and only if the consumption function is concave. Under this condition, effective risk aversion is increasing in wealth if and only if $\eta_t \leq \frac{1}{\phi-1}$, where ϕ denotes the ratio of apparent prudence to apparent risk aversion; a necessary condition, $\eta_t \leq \frac{1}{k-1}$, is shown to always hold. A parallel characterization is obtained for the concavity of effective risk aversion. Furthermore, effective risk aversion approaches zero at the borrowing constraint and $\frac{1}{k-1}$ as wealth grows without bound. In the special case where apparent prudence equals k times apparent risk aversion, effective risk aversion is always increasing and concave. In the Merton model, the canonical example of this special case, these properties hold whenever income exceeds subsistence consumption.

WORK IN PROGRESS

Household Problem with Risk Heterogeneity: A Continuous-Time Framework (2026)

Abstract: The classical framework assumes that the evolution of household welfare is deterministic and does not respond to risk sources, thus failing to distinguish the impact of aggregate risk versus idiosyncratic risk on household financial decisions. This paper relaxes this assumption, allowing the cost functional to have a stochastic response to any risk source. Using the Malliavin derivative as a sensitivity measure, the generalized HJB equation is derived through the Itô-Wentzell-Kunita formula. Compared to the classical framework, the generalized HJB equation includes a new term characterizing the cross-effect between household endowment risk exposure and welfare sensitivity to risk sources. We identify when the classical framework holds: risk sources do not directly affect household endowments (stochastic discounting, preference shocks, bequest functions, etc.), or households have no risk exposure (the Huggett model in continuous time).

COMPUTATIONAL WORK

Julia Implementation of "Financial Frictions and the Wealth Distribution" (2024)

Abstract: Independent implementation and replication of the model in "Financial Frictions and the Wealth Distribution, Fernández-Villaverde et al. (2023)" using Julia.

Open-Source Contribution for Modernization of ”Recursive Macroeconomic Theory” Codebase (2024) *Available at* GitHub Repository

Abstract: This repository contains Matlab solutions and updated functions from Ljungqvist & Sargent’s textbook (2018, MIT) for compatibility with Matlab 2024+.

EDUCATION

Shihezi University

2024-Present

Master of Finance (M.F.)

COMPUTATIONAL METHODS

Julia: Neural Networks, Numerical Analysis & PDEs, Tensor Calculation, Finite Difference Method

Matlab: DSGE

LANGUAGES

Chinese, English

Last updated: June 16, 2026